Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Friday 26 June 2026

1.1 Daily Limit

Trigonometric Limit

Evaluate the following limit:

$$\lim_{x\to 0} \frac{1-\cos(x)-\frac{1}{2}x^2}{x^4}$$

Solution: To evaluate the limit, we can use the Maclaurin series expansion for $\cos(x)$:

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \mathcal{O}(x^8)$$

Substitute this expansion into the numerator:

$$1-\cos(x)-\frac{1}{2}x^2 = 1 - \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \mathcal{O}(x^6)\right) - \frac{1}{2}x^2$$

Simplify the expression by distributing the negative sign and canceling terms:

$$= 1 - 1 + \frac{x^2}{2} - \frac{x^4}{24} + \mathcal{O}(x^6) - \frac{1}{2}x^2 = -\frac{x^4}{24} + \mathcal{O}(x^6)$$

Now, substitute the simplified numerator back into the limit:

$$\lim_{x\to 0} \frac{-\frac{x^4}{24} + \mathcal{O}(x^6)}{x^4} = \lim_{x\to 0}\left(-\frac{1}{24} + \mathcal{O}(x^2)\right)$$

As x approaches 0, the higher-order terms $\mathcal{O}(x^2)$ vanish. Thus, the limit evaluates exactly to:

$$-\frac{1}{24}$$

Saturday 27 June 2026

2.1 Daily Limit

Sequence Limit with Sine

Evaluate the limit:

$$\lim_{n\to\infty} \sin\left(\pi\sqrt{n^2+67^2}\right), \quad n \in \mathbb{N}$$

Solution: We want to find the limit of the sequence as $n \to \infty$. First, we extract n from the square root to expose its asymptotic behavior:

$$\sqrt{n^2+67^2} = n\sqrt{1+\frac{67^2}{n^2}}$$

Using the generalized binomial expansion $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$ for small x:

$$n\sqrt{1+\frac{67^2}{n^2}} = n\left(1+\frac{67^2}{2n^2}+\mathcal{O}\left(\frac{1}{n^4}\right)\right) = n+\frac{67^2}{2n}+\mathcal{O}\left(\frac{1}{n^3}\right)$$

Substitute this back into the argument of the sine function:

$$\sin\left(\pi\sqrt{n^2+67^2}\right) = \sin\left(\pi n+\frac{67^2\pi}{2n}+\mathcal{O}\left(\frac{1}{n^3}\right)\right)$$

Recall the trigonometric addition formula $\sin(A+B) = \sin(A)\cos(B)+\cos(A)\sin(B)$. For $A = \pi n$ and integer $n \in \mathbb{N}$, we know $\sin(\pi n) = 0$ and $\cos(\pi n) = (-1)^n$. Hence:

$$\sin\left(\pi n+\frac{67^2\pi}{2n}+\mathcal{O}\left(\frac{1}{n^3}\right)\right) = (-1)^n \sin\left(\frac{67^2\pi}{2n}+\mathcal{O}\left(\frac{1}{n^3}\right)\right)$$

Now take the limit as $n \to \infty$. The argument of the sine function approaches 0:

$$\lim_{n\to\infty}\frac{67^2\pi}{2n} = 0 \implies \lim_{n\to\infty}\sin\left(\frac{67^2\pi}{2n}\right) = 0$$

Since $(-1)^n$ alternates but remains bounded, the overall limit converges perfectly to 0:

$$0$$

Sunday 28 June 2026

3.1 Daily Limit

Repeated Integration Limit

For $x > 0$ and $n \geq 0$, let $F_0(x) = \ln(x)$ and

$$F_{n+1}(x) = \int_0^x F_n(t)\, dt.$$

Evaluate the limit:

$$\lim_{n\to\infty} \frac{n!F_n(1)}{\ln(n)}$$

Solution: We compute the first few iterations to establish a pattern for $F_n(x)$:

$$F_1(x) = \int_0^x \ln(t)\, dt = x\ln(x) - x$$
$$F_2(x) = \int_0^x (t\ln(t) - t)\, dt = \frac{x^2}{2}\ln(x) - \frac{x^2}{4} - \frac{x^2}{2} = \frac{x^2}{2}\ln(x) - \frac{3x^2}{4}$$

We hypothesize that $F_n(x)$ generally takes the form:

$$F_n(x) = \frac{x^n}{n!}\ln(x) - C_n x^n$$

Taking the derivative verifies the recurrence $\frac{d}{dx}F_n(x) = F_{n-1}(x)$:

$$F_n'(x) = \frac{nx^{n-1}}{n!}\ln(x) + \frac{x^{n-1}}{n!} - nC_n x^{n-1} = \frac{x^{n-1}}{(n-1)!}\ln(x) + \left(\frac{1}{n!} - nC_n\right)x^{n-1}$$

For this to exactly match $F_{n-1}(x) = \frac{x^{n-1}}{(n-1)!}\ln(x) - C_{n-1}x^{n-1}$, we equate coefficients:

$$\frac{1}{n!} - nC_n = -C_{n-1} \implies nC_n = C_{n-1} + \frac{1}{n!} \implies n!C_n = (n-1)!C_{n-1} + \frac{1}{n}$$

Let $A_n = n!C_n$. The sequence simplifies to the basic recurrence $A_n = A_{n-1} + \frac{1}{n}$. Since $F_0(x) = \ln(x)$, $A_0 = 0$. Thus A_n is the n-th Harmonic number, $H_n = \sum_{k=1}^n \frac{1}{k}$, giving $C_n = \frac{H_n}{n!}$. The general formula is:

$$F_n(x) = \frac{x^n}{n!}(\ln(x) - H_n)$$

Evaluate at $x = 1$:

$$F_n(1) = \frac{1^n}{n!}(\ln(1) - H_n) = -\frac{H_n}{n!}$$

Substitute this back into the required limit:

$$\lim_{n\to\infty} \frac{n!\left(-\frac{H_n}{n!}\right)}{\ln(n)} = \lim_{n\to\infty} \frac{-H_n}{\ln(n)}$$

We invoke the asymptotic expansion of the harmonic numbers $H_n \sim \ln(n) + \gamma$, where γ is the Euler-Mascheroni constant:

$$\lim_{n\to\infty} \frac{-(\ln(n)+\gamma)}{\ln(n)} = \lim_{n\to\infty}\left(-1 - \frac{\gamma}{\ln(n)}\right) = -1$$

Monday 29 June 2026

4.1 Daily Limit

Series Limit

Evaluate the limit:

$$\lim_{n\to\infty} n\left(\sum_{k=0}^{\infty}\frac{k}{n^k}\right)$$

Solution: First, we evaluate the closed form of the infinite series $S = \sum_{k=0}^{\infty}\frac{k}{n^k}$. Recall the formula for an infinite geometric series where $|x| < 1$:

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$$

Differentiating both sides with respect to x yields:

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}$$

Multiply both sides by x to match the algebraic structure of our sum:

$$\sum_{k=1}^{\infty} kx^k = \frac{x}{(1-x)^2}$$

Since the $k = 0$ term contributes exactly 0 to the sum, we rewrite our original series by substituting $x = \frac{1}{n}$. Since $n \to \infty$, we assume $n > 1$ so the condition $|x| < 1$ is satisfied:

$$\sum_{k=0}^{\infty} k\left(\frac{1}{n}\right)^k = \frac{\frac{1}{n}}{(1-\frac{1}{n})^2} = \frac{\frac{1}{n}}{(\frac{n-1}{n})^2} = \frac{n}{(n-1)^2}$$

Now, substitute the evaluated closed form back into the limit expression:

$$\lim_{n\to\infty} n\left(\frac{n}{(n-1)^2}\right) = \lim_{n\to\infty}\frac{n^2}{n^2-2n+1}$$

Divide the numerator and the denominator by the highest degree term, n^2:

$$\lim_{n\to\infty}\frac{1}{1-\frac{2}{n}+\frac{1}{n^2}} = \frac{1}{1-0+0} = 1$$

Tuesday 30 June 2026

5.1 Daily Limit

Limit with Exponentials and Trigonometry

Evaluate the limit:

$$\lim_{x\to 0} \frac{e^x - \cos(x) - x - \frac{x^2}{2}}{x^2}$$

Solution: We can solve this limit by exploiting Taylor series expansions at $x = 0$, which naturally tracks non-vanishing terms. The Maclaurin series for e^x and $\cos(x)$ are given by:

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)$$

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \mathcal{O}(x^6)$$

Substitute these expansions into the numerator of our limit:

$$N(x) = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)\right) - \left(1 - \frac{x^2}{2} + \mathcal{O}(x^4)\right) - x - \frac{x^2}{2}$$

Group the terms by their respective powers of x:

$$N(x) = (1-1) + (x - x) + \left(\frac{x^2}{2} - \left(-\frac{x^2}{2}\right) - \frac{x^2}{2}\right) + \frac{x^3}{6} + \mathcal{O}(x^4)$$

Simplifying the quadratic x^2 coefficient specifically:

$$\frac{1}{2} + \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$$

The numerator consolidates completely to:

$$N(x) = \frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)$$

Now evaluate the limit by dividing by the denominator x^2:

$$\lim_{x\to 0} \frac{\frac{x^2}{2} + \frac{x^3}{6} + \mathcal{O}(x^4)}{x^2} = \lim_{x\to 0} \left(\frac{1}{2} + \frac{x}{6} + \mathcal{O}(x^2)\right)$$

Taking $x \to 0$, the x and all higher-order terms evaluate to 0, directly leaving:

$$\frac{1}{2}$$

Alternatively, employing L'Hôpital's Rule consecutively since the limit is in a $0/0$ *indeterminate form:*

$$\lim_{x\to 0} \frac{e^x - \cos(x) - x - \frac{x^2}{2}}{x^2} \stackrel{\text{L'H}}{=} \lim_{x\to 0} \frac{e^x + \sin(x) - 1 - x}{2x}$$

This is still $0/0$, so apply L'Hôpital's Rule a second time:

$$\stackrel{\text{L'H}}{=} \lim_{x\to 0} \frac{e^x + \cos(x) - 1}{2} = \frac{1 + 1 - 1}{2} = \frac{1}{2}$$